# ENGR 290 Midterm Study Guide

Materials and Thermodynamics • Steve Ferguson • November 9, 2026

## What to bring

The midterm information file allows 110 minutes, a basic scientific calculator and one handwritten, double-sided 8.5 × 11 inch sheet. Print your name and student number at the top and submit the sheet with the exam. Phones and graphing calculators are excluded. The posted file retains a 2025 heading. The revised 2026 announcement gives Monday, November 9.

## Where to start

If the question gives an alloy composition and temperature, draw the vertical composition line first. If it asks how much of each phase forms, add a horizontal tie line. Practice these two moves on the Al–Si figure in the test before reading the solution.

If the question gives a diffusion depth, identify the surface, initial and target concentrations. If it describes a polymer, identify the chain structure and where the temperature sits relative to the glass transition.

## Match the guide to the assignments

Assignment 1 Q1–6 → density, cubic geometry, planes, vacancy fraction and carburizing. Assignment 2 Q1–4 → polymers and solidification. Q5–11 → phase diagrams, cooling curves and alloy additions. Q12–14 → polymer repeat units, service life and ABS.

Ferguson’s midterm study list

Exam information sheet

Course announcements and lecture corrections

# Atomic structure and unit cells

## Atoms and electrons

Atomic number counts protons. a neutral atom has the same number of electrons. Valence electrons participate in bonding. An orbital holds at most two electrons with opposite spins. For oxygen, $1s^2\ 2s^2\ 2p^4$ accounts for all eight electrons.

## From a lattice to a material

A lattice repeats through space. a crystal structure combines a lattice with its atomic basis. A unit cell describes the repeating geometry. A grain is a crystal region with a common orientation, and grain boundaries separate differently oriented grains. Amorphous solids lack long-range periodic order, although short-range order can remain.

## Count shared atoms

A corner contributes 1/8 of an atom, a face contributes 1/2, and a body-centred atom contributes 1. Thus simple cubic has 1 atom per conventional cell, BCC has 2, and FCC has 4. For identical hard spheres: SC has $a = 2r$. BCC has $\sqrt{3}a = 4r$. FCC has $\sqrt{2}a = 4r$. Their coordination numbers are 6, 8, and 12 respectively.

## Density and units

For a monatomic cubic crystal, $\rho = nM/(N_a a^3)$. Here n is atoms per cell, M is molar mass, and $N_a$ is Avogadro's constant. Use a in cm with M in g/mol to obtain $g/cm^3$. For a macroscopic sample use $\rho = m/V$ directly. A mole-sized cube is not a unit cell.

## Worked example

One mole of MgO has mass 24.305 + 15.999 = 40.304 g. For the assignment cube, 22.37 mm = 2.237 cm. $V = 2.237^3 = 11.1943\ cm^3$, so $\rho = 40.304/11.1943 = 3.600\ g/cm^3$. Always cube the converted length.

$$\rho = \frac{nM}{N_a a^3}$$

FCC packing factor $= 4(4\pi r^3/3)/a^3 = \pi/(3\sqrt{2}) \approx 0.74$. Face diagonal contact gives $\sqrt{2}a = 4r$. This is the corrected Week 2a result.

# Directions and Miller indices

## Directions use square brackets

Subtract start coordinates from end coordinates in lattice units, clear fractions, and reduce to the smallest integer ratio. From (0,0,0) to (1/2,1/2,0), the direction is [110]. A negative component is conventionally written with an overbar. A family of equivalent directions uses angle brackets ⟨uvw⟩.

## Planes use parentheses

Choose a plane that does not pass through the origin, find intercepts in units of a, b, c, take reciprocals, then clear fractions. An infinite intercept gives index zero. Intercepts a, 2b, ∞ give reciprocals 1, 1/2, 0 and indices (210). Braces {hkl} identify a family of planes.

## Reverse the process

The conventional intercepts of (hkl) are a/h, b/k, c/l. A zero index means the plane is parallel to that axis. In a cubic crystal [hkl] is normal to (hkl). This simple identification does not generally hold in other crystal systems.

## Cubic angles and spacing

For cubic directions, $\cos\theta = (uu' + vv' + ww')/[\sqrt{(u^2+v^2+w^2)}\,\sqrt{(u'^2+v'^2+w'^2)}]$. For cubic plane spacing, $d_{hkl} = a/\sqrt{(h^2+k^2+l^2)}$. Set the calculator to degrees when reporting angles.

## Worked example

For [110] and [111], $\cos\theta = 2/\sqrt{6}$, so $\theta = 35.3°$. Assignment 1 Q4 has a negative second index. For (3 −1 1), the intercepts are a/3, −b, c.

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

In hexagonal plane notation (h k i l), i = −(h + k). For cubic cells, a displacement between fractional coordinates has physical length $a\sqrt{(\Delta u^2 + \Delta v^2 + \Delta w^2)}$.

# Bonding and point defects

## Primary and secondary bonds

Ionic bonding involves electrostatic attraction between oppositely charged ions. Covalent bonding shares electrons and is directional. Metallic bonding has delocalized electrons that help explain electrical conductivity and non-directional bonding. Secondary interactions include dipole attractions and hydrogen bonding. Bonding helps explain properties, but crystal structure and defects also matter.

## Classify the defect

A vacancy is an empty regular site. A substitutional atom replaces a host atom. an interstitial occupies a gap. A self-interstitial is a host atom in such a gap. A Frenkel defect pairs a vacancy with an interstitial of the same ion. A Schottky defect removes ions in a charge-neutral stoichiometric combination. In a 1:1 ionic solid this is one cation and one anion vacancy.

## Vacancies follow an Arrhenius relation

Write vacancy fraction $f = C\exp(-E_v/kT)$, where $C$ is a prefactor. With constant $C$, $f_2/f_1 = \exp[(E_v/k)(1/T_1 - 1/T_2)]$. Use absolute temperature in kelvin when k is in eV/K, and match the energy units. A plot of ln f against $1/T$ has slope $-E_v/k$.

## Reason before calculating

If temperature rises, equilibrium vacancy fraction rises. If your result falls, check the sign of the exponent. When an initial fraction is supplied, use the ratio form. Setting $C = 1$ unnecessarily can discard the given calibration.

## Worked example

Using the assignment values, $T_1 = 673.15$ K, $T_2 = 933.15$ K, $E_v = 0.76$ eV, $k = 8.62 \times 10^{-5}$ eV/K and $f_1 = 2.29 \times 10^{-5}$: $f_2 = f_1 \exp[(0.76/k)(1/673.15 - 1/933.15)] \approx 8.80 \times 10^{-4}$. This is a fraction, or about 0.0880%.

$$\ln(f_2/f_1) = \frac{E_v}{k}(1/T_1 - 1/T_2)$$

A compound has a fixed stoichiometry. A solid solution can vary in composition within its solubility limits. An element that takes different crystal forms exhibits allotropy.

# Dislocations and strengthening

## Line defects enable slip

An edge dislocation terminates an extra half-plane of atoms. Its Burgers vector is perpendicular to the line. For a screw dislocation, the Burgers vector is parallel to the line. A mixed dislocation has both characters. Motion of dislocations produces permanent slip without requiring an entire plane of bonds to break at once.

## Slip systems and repeat distances

A slip system combines a slip plane and a direction within it. FCC commonly slips on {111} in ⟨110⟩ directions. A perfect FCC Burgers vector is $(a/2)\langle 110\rangle$ with magnitude $a/\sqrt{2}$. Distinguish a direction such as [110] from a vector with a physical length.

## Resolve the applied stress

Schmid's law is $\tau = \sigma \cos\varphi \cos\lambda$. $\varphi$ is the angle between the loading axis and the slip-plane normal. $\lambda$ is the angle between the loading axis and slip direction. Slip begins when $|\tau|$ reaches the critical resolved shear stress. For a valid slip system the Schmid factor is at most 0.5.

## Obstacles increase strength

Cold working increases dislocation density, making dislocations obstruct each other. Solute atoms create strain fields. Grain boundaries and particles impede motion. Within the usual Hall–Petch regime, $\sigma_Y = \sigma_0 + k_Y/\sqrt{d}$. Smaller grains increase the grain-size contribution. keep d consistent with the units of $k_Y$.

## Worked example

If $\sigma = 120$ MPa, $\varphi = 60°$ and $\lambda = 30°$, $\tau = 120(0.5)(0.8660) = 51.96$ MPa. If the critical resolved shear stress is 60 MPa, this system has not begun to slip. Its required applied stress is $60/(0.5 \times 0.8660) = 138.6$ MPa.

$$\tau = \sigma \cos\varphi \cos\lambda$$

A stacking fault interrupts a layer sequence. A twin mirrors the lattice across a plane. A small-angle tilt boundary can be an array of edge dislocations. Screw segments can cross-slip onto another plane containing the same Burgers vector.

# Diffusion and assignment practice

## How atoms move

Vacancy diffusion involves an atom jumping into a neighbouring vacant site. Interstitial diffusion moves atoms between interstitial positions. Diffusivity commonly follows $D = D_0 \exp(-Q/RT)$ for molar activation energy Q. Use k instead of R for energy per atom. Increased temperature generally increases D.

## Steady diffusion

Fick's first law is $J = -D\, dC/dx$. The minus sign indicates transport down the concentration gradient. With constant D and a linear profile through a slab, $J = -D(C_2 - C_1)/L$. If C is in kg/m$^3$, flux is kg/(m$^2$·s).

## A changing concentration profile

For a semi-infinite solid with uniform initial concentration $C_0$, constant surface concentration $C_s$, and constant D: $(C(x,t) - C_0)/(C_s - C_0) = 1 - \text{erf}[z]$, where $z = x/(2\sqrt{(Dt)})$. Find z from an error-function table, then $t = x^2/(4Dz^2)$. This model assumes a sufficiently thick body so the far boundary has no influence.

## Read the table carefully

An erf table and an erfc table are different: $\text{erfc}(z) = 1 - \text{erf}(z)$. Normalize concentrations first, then select the matching table value. For fixed normalized concentration and D, time scales with depth squared. Convert millimetres to metres before using D in m$^2$/s.

## Worked example

Assignment-style carburizing: $C_s$ = 1.0 wt %, $C_0$ = 0.2 wt %, target C = 0.6 wt %, x = 0.001 m, and D = 2.98 × $10^{-11}$ m$^2$/s. The normalized concentration is 0.5, so erf(z) = 0.5 and z ≈ 0.4769. Thus $t = 0.001^2/[4(2.98 \times 10^{-11})(0.4769)^2] \approx 3.69 \times 10^4$ s ≈ 10.25 h.

$$z = \frac{x}{2\sqrt{Dt}}$$

With constant D, Fick's second law gives $\partial C/\partial t = D\, \partial^2 C/\partial x^2$. At the same case depth and concentration, $D_1 t_1 = D_2 t_2$. Compare hourly furnace rate × time to decide which temperature costs less.

# Nucleation and growth

### What makes an embryo survive?

Below the freezing temperature, forming solid lowers bulk free energy. Creating its surface costs energy. A tiny embryo has too much surface per unit volume and tends to dissolve. Above the critical radius, further growth lowers total free energy.

$$\Delta G = 4\pi r^2 \gamma + \left(4\pi r^3 / 3\right) \Delta G_v$$

Here $\gamma$ is positive and $\Delta G_v$ is negative. Setting the slope to zero gives $r^* = 2\gamma / |\Delta G_v|$. Greater undercooling increases the driving force and decreases $r^*$. A solid precipitate forming inside another solid also pays an elastic strain-energy cost.

### Explain what the cooling curve does

Removing sensible heat lowers temperature ($Q = mc\Delta T$). Freezing releases latent heat ($Q = mL$). During equilibrium freezing of a pure material, heat can leave while temperature stays nearly constant. An alloy freezing through a range typically changes slope instead.

### Why can the interface stay flat?

Heat leaves through the solid. When the liquid ahead is warmer, a tip projecting into it stops advancing or remelts. The rest of the front catches up. In an undercooled melt the tip can instead grow ahead as a dendrite.

### Choose a structure for a turbine blade

Homogeneous nuclei form within the liquid. A mould wall, grain boundary or seed lowers the nucleation barrier. Many nuclei generally give finer grains. Directional heat removal encourages columnar grains. Selecting one growing grain produces a single crystal and removes grain-boundary sliding as a high-temperature failure mechanism.

# Polymer chains and temperature

## Choose among the three classes

Thermoplastics have linear or branched chains. Heating permits chain motion and reshaping. Thermosets form dense permanent covalent networks during cure and resist flow on reheating. Elastomers have flexible chains joined by sparse cross-links. Above their glass transition, long segments uncoil under load and recoil on unloading.

## Read modulus and cooling curves

E is the initial slope of stress against strain. Below Tg, amorphous segments are glassy and stiff. Across Tg segmental motion increases and E falls. A semicrystalline polymer can retain stiffness above Tg because crystallites persist until Tm. On a specific-volume versus temperature graph, Tg changes slope. Melting crystalline material produces a volume change and involves latent heat.

## Explain drawing and ABS

Drawing straightens and aligns thermoplastic chains along the load. More load then acts along strong covalent backbones, increasing strength in that direction. In ABS, increasing the rubbery butadiene-containing fraction favours impact absorption and ductility. Increasing rigid SAN generally favours strength and stiffness.

## Choose a forming process

Extrusion forces softened polymer through a die to make continuous pipe or strip of constant cross-section. Injection molding fills a closed mould, cools and ejects a discrete part. Sulfur vulcanization forms cross-links in rubber. A plasticizer increases segment mobility.

# Polymer calculations and reactions

## Chain length versus cell contents

Degree of polymerization counts repeat units per chain. Divide chain molar mass by repeat-unit molar mass, using the same units. For a distribution, number-average molar mass is $\Sigma N_i M_i / \Sigma N_i$ and weight-average is $\Sigma N_i M_i^2 / \Sigma N_i M_i$. Longer chains receive more weight in the second average.

$$DP = M / M_r \quad n(cell) = \rho V N_a / M_r$$

Assignment 2 Q12 uses V = 1.450 × 0.569 × 0.740 $nm^3$. Multiply by $10^{-21}$ to obtain $cm^3$. The cell mass is 0.89V grams. Dividing by $42/N_a$ grams per repeat gives 7.79, consistent with eight repeats per cell.

## Estimate service life at constant strain

$$\sigma(t) = \sigma_0 \exp(-t/\lambda)$$

For Assignment 2 Q13, λ = −100/ln(2400/2500) = 2450 h. At the 2100 psi threshold, t = −2450 ln(2100/2500) = 427 h from loading, or 327 h after the observation. Constant stress and increasing strain describes creep.

## Identify a polymerization mechanism

In addition polymerization, a radical can open a double bond and successive monomers add without losing a small molecule. Chains terminate by combination or disproportionation. Condensation polymerization often releases a small molecule such as water. Polyethylene's repeat is $–CH_2–CH_2–$. Do not count two carbon atoms as two repeats.

## Lecture extensions

An adhesive can set by chemical cure, cooling a hot melt, or solvent loss. Electronic electroactive polymers typically respond rapidly at high electric fields. Ionic actuators can bend at low voltage but may require electrolyte and respond more slowly.

# Reading a phase diagram

## First locate the point

A phase is homogeneous in structure and composition at equilibrium. Two grains can belong to one phase. A component is an independent chemical constituent. On a binary diagram identify the solute and the axis units before locating overall composition $C_0$ and temperature. Use the at.% scale for an at.% question.

## Read the boundaries

Cool vertically at fixed $C_0$. Crossing the liquidus produces the first solid. Draw a horizontal tie line to the solid boundary to find its composition. At the solidus the last liquid disappears. The solvus gives a solid solubility limit. Freezing range = liquidus temperature − solidus temperature.

## Read amounts from opposite arms

$$w(\text{¿¿}) = (C_r - C_0)/(C_r - C)\, w(\text{¿}) = 1 - w(\text{¿¿})$$

On a wt.% axis those are mass fractions. A 30 kg alloy at 30 wt.% B with tie-line ends 10 and 70 wt.% B has 20 kg of the left phase and 10 kg of the right. Check the solute balance: 20(0.10) + 10(0.70) = 9 kg.

## Count degrees of freedom

F = C − P + 2 includes pressure. At fixed pressure F = C − P + 1. For a binary, one phase therefore gives 3 in the full convention or 2 at fixed pressure. State your convention. Assignment 2 Q5 prints 3, 2, 3. At 500 °C and 20 wt.% Cu the supplied equilibrium diagram shows two solid phases, giving 2 under the full rule. Identify the field when explaining the discrepancy.

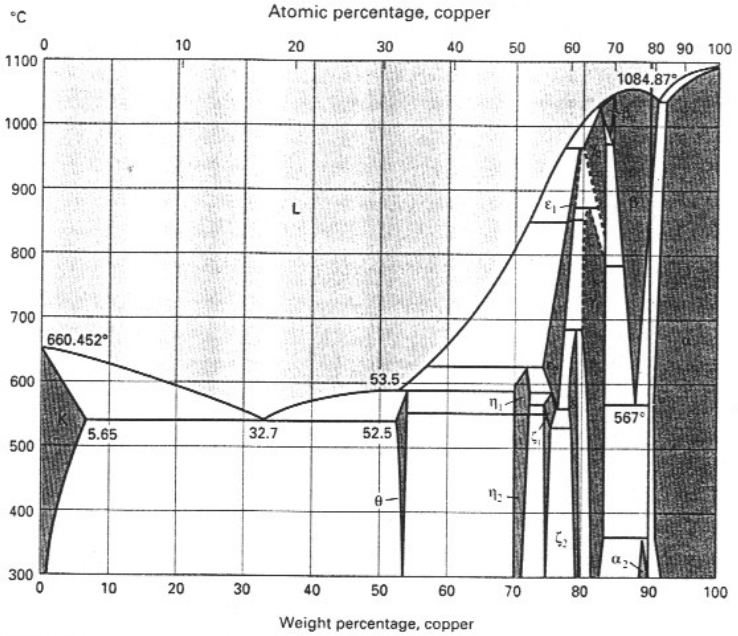



FIGURE 5.5-9 Al–Cu phase diagram. (After *Binary Alloy Phase Diagrams*, Vol. 1, T. B. Massalski, ed., American Society for Metals, Metals Park, Ohio, 1986.)

Assignment 2, Q5. Start at 20 wt.% Cu on the lower axis. At 500 °C the point lies below the eutectic horizontal, between the Al-rich solid and the intermediate compound.

# Reactions and steel microstructures

### Find the horizontal line and read above and below

On cooling, eutectic is $L \rightarrow \alpha + \beta$. Eutectoid is $\gamma \rightarrow \alpha + \beta$. Peritectic is $L + \alpha \rightarrow \beta$. Peritectoid is $\alpha + \beta \rightarrow \gamma$. Monotectic is $L_1 \rightarrow L_2 + \alpha$. A congruently melting compound produces liquid of the same composition. A metastable phase can persist because transformation to the lower-energy state is kinetically slow.

### Follow an off-eutectic alloy

In the Week 5B Pb–Sn example, 40 wt.% Sn lies left of 61.9 wt.% Sn. Primary $\alpha$ forms first. At 183 °C the remaining liquid forms eutectic $\alpha + \beta$. Just above that temperature the tie line runs from 19.2 to 61.9 wt.% Sn. The liquid fraction $(40 - 19.2)/(61.9 - 19.2) \approx 0.487$ becomes the eutectic microconstituent. Primary $\alpha$ plus $\alpha$ in the eutectic make up total $\alpha$.

### Use Ferguson's steel values

At 723 °C and 0.8 wt.% C, austenite ($\gamma$, FCC) transforms into pearlite, alternating ferrite ($\alpha$, BCC) and cementite ($Fe_3C$, 6.67 wt.% C). Pearlite contains two phases. High-temperature $\delta$ ferrite is also BCC. The Fe–$Fe_3C$ diagram treats cementite as metastable relative to graphite.

For slow cooling of 0.4 wt.% C steel, proeutectoid ferrite forms first. Remaining $\gamma$ becomes pearlite. At 1.2 wt.% C the first constituent is proeutectoid cementite, followed by pearlite. At exactly 0.8 wt.% C all $\gamma$ becomes pearlite.

# Mixtures and additional lecture checks

## How much pure solute must be added?

For an initial mass m and solute fraction $c_0$, read the required target fraction c from the appropriate boundary. Write the balance before substituting.

$$(mc_0 + x)/(m + x) = c \Longrightarrow x = m(c - c_0)/(1 - c)$$

For 100 kg at 20 wt.% solute, reaching 50 wt.% takes 60 kg of pure solute. Check that 80 kg solute in 160 kg gives 50%. If the material added is itself an alloy, replace x in the numerator with its solute fraction times x. To convert wt.% to at.%, divide each component mass by its molar mass and normalize the mole amounts.

## Atomic arrangement and solid solubility

Principal, orbital, magnetic and spin quantum numbers specify shell, subshell, orbital orientation and spin. An orbital holds two opposite spins. Seven crystal systems describe unit-cell geometry and fourteen Bravais lattices include centring. For extensive substitutional solubility look for similar radii (about 15% or less difference), structure, electronegativity and valence. Small atoms such as carbon can occupy interstitial sites.

## Use a one-component pressure–temperature diagram

A single field has two independent variables under the full phase rule. A two-phase boundary has one. The triple point has none. Follow a constant-pressure path to identify the order of phase changes. The melting boundary for ordinary ice slopes towards lower temperature as pressure rises because liquid water occupies less volume.

## What to copy to the handwritten sheet

Front: cubic geometry, direction and plane brackets, Burgers vector, Schmid factor, vacancy ratio, both Fick laws and the erf argument. Back: a labelled tie line, the five reactions, the steel eutectoid, mass balance, polymer repeat-unit count and relaxation. Beside each equation, write the required units and what the answer counts.